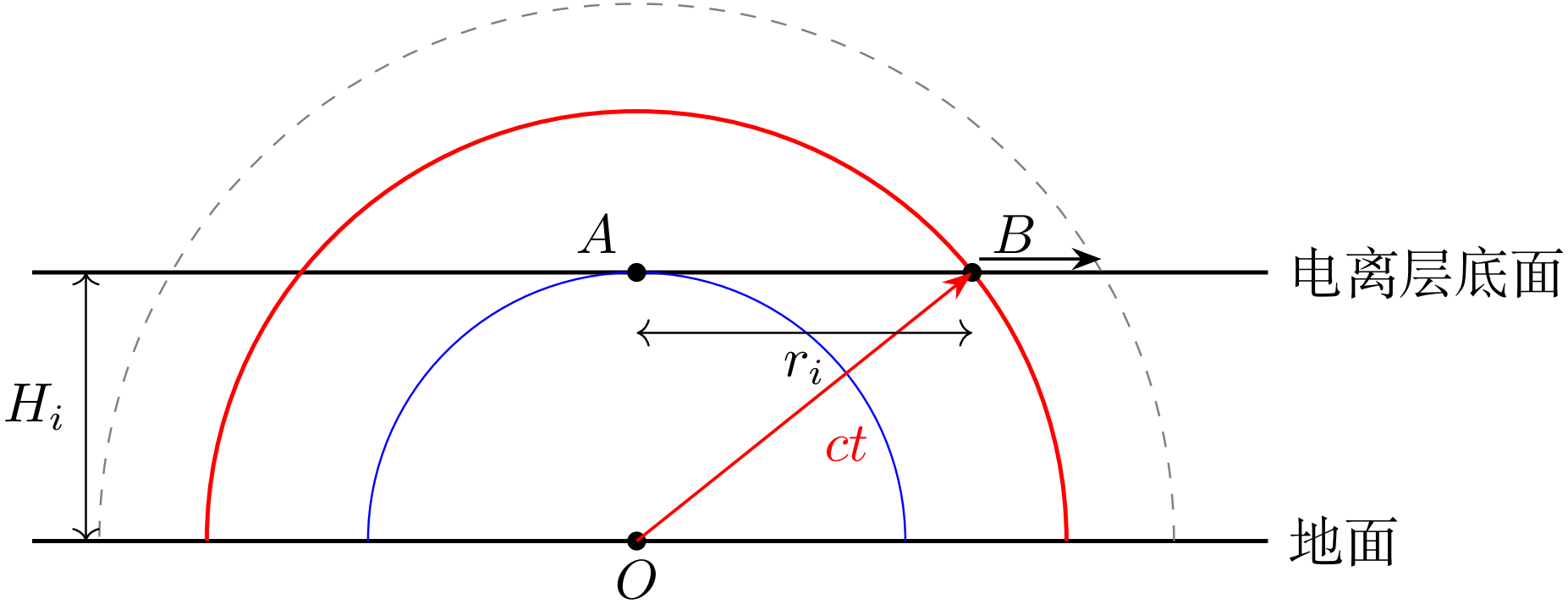
球面波以 c 传播，交线的径向速度可大于 c



$$r_i^2 + H_i^2 = c^2 t^2$$